

played a hand in the development of nanotechnology with this theory as well as with the method he devised of calculating the size of particles by determining the light they scatter.

First Electron Microscope (1931)

Working with Max Knott, the German Scientist Ernst Ruska developed the world's first Electron Microscope that would enable us to recognise object that were smaller than the wavelength of light. It transcended the wavelength of visible light and exceeded the detail and clarity of the traditional light microscope. This was an important step in the development of techniques and instrumentation that would eventually enable research in the nanoscale.



World's First Electron Microscope

The Transistor (1947)

Until the mid-1940's, the electronic world depended on vacuum tubes for almost everything - Converting AC to DC, Amplifying an electronic signal, Switching telephone calls, Building the high-speed computer, ENIAC. However their bulky yet fragile nature as well as their tendency to overheat, was a hindrance to the electronics world. In 1947 William Shockley, Walter Brattain and John Bardeen working at Bell Labs created the world's first point-contact transistor. They went on to receive the Nobel Prize in physics and their invention of the transistor and the integrated circuit marked the beginning of microelectronics, a field that relies on tools for miniaturisation. The semiconductor industry is one of the largest technology drivers in the field of nanotechnology.



World's First Transistor

Mondisperse Colloidal Materials (1950)

Victor La Mer and Robert Dinegar developed the theory and a process for growing monodisperse colloidal materials. Controlled ability to fabricate colloids enables myriad industrial uses such as specialised papers, paints, and thin films, even dialysis treatments.